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CABINET AND SERVER

Abstract

A cabinet comprises a chassis, a bracket, and a self-locking component. The bracket comprises a support and a handle, the support is movable between a work position and a transition position and the support is further rotatable between the transition position and an open position relative to the chassis, the handle is rotatable between a handle-lifted-up position and a handle-pressed-down position relative to the support. The self-locking component is connected to the support and the chassis. When the handle rotates from the handle-pressed-down position to the handle-lifted-up position, the support moves from the work position to the transition position; when the support rotates from the transition position to the open position, the support extends out of the chassis, the self-locking component is configured for locking the support at the open position, to expose a space under the bracket. A server with the cabinet is also disclosed.

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Background/Summary

FIELD

[0001] The disclosure herein generally relates to information computing systems, and more particularly relates to a cabinet and a server.

BACKGROUND

[0002] Server usually has two layers of spaces to install electronic devices, and electronic devices in each layer sometimes need to be removed for maintenance or replacement, however, the traditional way of installing and removing the electronic devices is inconvenient, resulting in low efficiency.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Many aspects of the disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

[0004] FIG. **1** is an isometric view of an embodiment of a cabinet of a server according to the present disclosure, showing the cabinet in a working state.

[0005] FIG. **2** is an isometric view of the cabinet in FIG. **1**, showing the cabinet in an open state.

[0006] FIG. **3** is a side view of the cabinet in FIG. **1**, showing the cabinet in the working state.

[0007] FIG. **4** is a side view of the cabinet in FIG. **2**, showing the cabinet in the open state.

[0008] FIG. **5** is an exploded view of the cabinet in FIG. **3**, showing a first sheet disassembled from the cabinet.

[0009] FIG. **6** is a side view of a part of the cabinet in FIG. **3** and the first sheet in FIG. **5**, showing a support on a work position and a handle on a handle-pressed-down position.

[0010] FIG. **7** is a side view of the part of the cabinet in FIG. **6** and the first sheet in FIG. **5**, showing the handle on a handle-lifted-up position.

[0011] FIG. **8** is a side view of the part of the cabinet in FIG. **7** and the first sheet in FIG. **5**, showing the support on a transition position.

[0012] FIG. **9** is a side view of the part of the cabinet in FIG. **8** and the first sheet in FIG. **5**, showing the support rotated to an open position.

[0013] FIG. **10** is a side view of a part of the cabinet and a second sheet in FIG. **3**, showing a support at a work position.

[0014] FIG. **11** is a side view of the part of the cabinet and the second sheet in FIG. **10**, showing the support at a transition position.

[0015] FIG. **12** is a side view of the part of the cabinet and the second sheet in FIG. **11**, showing the support rotated to an open position.

[0016] FIG. **13** is an isometric view of an embodiment of a self-locking component of according to the present disclosure.

[0017] FIG. **14** is an isometric view of the self-locking component in FIG. **13**, showing a third pin positioned in a limiting hole.

[0018] FIG. **15** is an exploded view of the self-locking component in FIG. **14**.

DETAILED DESCRIPTION

[0019] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, baffle structures, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts have been exaggerated to better illustrate details and features of the present disclosure.
[0020] The present disclosure, including the accompanying drawings, is illustrated by way of examples and not by way of limitation. Several definitions that apply throughout this disclosure will now be presented. It should be noted that references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and such references mean “at least one”.
[0021] The term “comprising” means “including, but not necessarily limited to;” it specifically indicates open-ended inclusion or membership in a so-described combination, group, series, and the like.
[0022] Without a given definition otherwise, all terms used have the same meaning as commonly understood by those skilled in the art. The terms used herein in the description of the present disclosure are for the purpose of describing specific embodiments only, and are not intended to limit the present disclosure.
[0023] As shown in FIG. **1** to FIG. **4**, a server **200** in an embodiment includes a cabinet **100** and a plurality of electronic devices. The plurality of electronic devices are installed in the cabinet **100**, and the plurality of electronic devices can be removed from the cabinet **100**. The cabinet **100** has a lower-layer space and an upper-layer space within, the upper-layer space is above the lower-layer space. Some of the plurality of electronic devices are placed in the lower-layer space and others of the plurality of electronic devices are placed in the upper-layer space. For example, electronic devices can be motherboards, hard drives, expansion cards and so on.
[0024] The cabinet **100** includes a chassis **10**, a bracket **20**, and a self-locking component **30**. The chassis **10** is a part of the shell of the server **200**, and the lower-layer space and the upper-layer space are inside the chassis **10**. The chassis **10** is used to hold electronic devices in the lower-layer space. The bracket **20** includes a handle **21** and a support **22**, as shown in FIG. **5** to FIG. **9**, the support **22** is used to hold electronic devices in the upper-layer space. The support **22** is movable relative to the chassis **10** between a work position and a transition position, FIGS. **3**, **6** and **10** show that the support **22** is at the work position and FIGS. **8** and **11** show that the support **22** is at the transition position. The support **22** is also rotatable relative to the chassis **10** between a transition position and an open position, FIGS. **4** and **14** show that the support **22** is at the open position. The self-locking component **30** connects the chassis **10** and the support **22**, the self-locking component **30** is used to lock the support **22** at the open position.
[0025] The support **22** between the work position and the transition position is in the upper-layer space, but the support **22** at the open position extends out of the chassis **10**, to expose the lower-layer space, so that it is more convenient to remove or install the electronic devices in the lower-layer space, because if the support **22** stays in the upper-layer space, the lower-layer space is covered by the support **22**, and it is inconvenient to remove or install the electronic devices in the lower-layer space.
[0026] In some embodiments, the support **22** is used to fix a motherboard. When the support **22** is at the work position, the motherboard is plugged into a socket of the server **200**. When the support **22** is at the transition position, the motherboard is pulled out of the socket of the server **200**. When the support **22** is at the open position, the support **22** lifts the motherboard out of the chassis **10**, so the motherboard can be removes or installed easier.

[0027] As shown in all figures, the support **22** moves in a direction X between the work position and the transition position, the support **22** rotates around a direction Y between the transition position and the open position, the lower-layer space and the upper-layer space are arranged in a direction Z, the direction X and the direction Y is horizontal, the direction Z is vertical.
[0028] As shown in FIG. **5** to FIG. **9**, the handle **21** is used to be held by users. The handle **21** is rotatably installed on the support **22**. The handle **21** is rotatable between a handle-lifted-up position and a handle-pressed-down position relative to the support **22**.
[0029] As shown in FIG. **2** and FIG. **5**, in some embodiments, the chassis **10** has two first sheets **11** and two second sheets **12**, the two first sheets **11** are fixed on the two opposite side of the chassis **10** in the direction Y, the two second sheets **12** are also fixed on the two opposite side of the chassis **10** in the direction Y, and the first sheet **11** and the second sheet **12** on the same side are arranged in the direction X. Each first sheet **11** has a bended sheet **11***a* on the bottom, the two bended sheets **11***a* are in the same horizontal plane, the two bended sheets **11***a* are used to support the support **22**, so that the support **22** will not drop down to the lower-layer space by gravity, and the support **22** slides on the two bended sheets **11***a* between the work position and the transition position.
[0030] As shown in FIG. **5** to FIG. **9**, the handle **21** has two first pins **211** on two opposite sides in the direction Y, the support **22** has two arc grooves **221** on two opposite sides in the direction Y, the first pin **211** extends through the arc groove **221** on the same side, and the first pin **211** is slidable in the arc groove **221**, when the handle **21** arrives at handle-lifted-up position, the first pin **211** is on a top end of the arc groove **221**, and when the handle **21** arrives at handle-pressed-down position, the first pin **211** is on a bottom end of the arc groove **221**.
[0031] As shown in FIG. **5** to FIG. **9**, each first sheet **11** of the chassis **10** has a first sliding groove **111**. The first pin **211** is also slidable into or out of the first sliding groove **111** on the same side. The first sliding groove **111** includes a first segment **111***a*, a second segment **111***b*, and a third segment **111***c*, the first segment **111***a* and the third segment **111***c* extend in the direction Z, the second segment **111***b* extends in the direction X, the left end of the second segment **111***b* is communicated to the top end of the first segment **111***a*, the right end of the second segment **111***b* is communicated to the bottom end of the third segment **111***c*, the top end of the third segment **111***c* extends through the first sheet **11** to allow the first pin **211** to slide into or out of the first sliding groove **111**.
[0032] When the motherboard is plugged into the socket, the support **22** is at the work position and the handle **21** is at the handle-pressed-down position, the first pin **211** is on a bottom end of the first segment **111***a*. When users rotate up the handle **21** from the handle-pressed-down position to the handle-lifted-up position (from FIG. **6** to FIG. **7**), because the first pin **211** is limited in the first segment **111***a* and the arc groove **221** simultaneously, the first pin **211** pushes against the left-side of the first segment **111***a* shown in FIG. **6** and FIG. **7** during the rotation of the handle **21**, so that the support **22** is moved towards the transition position, to disconnect the motherboard from the socket; when the handle **21** arrives at handle-lifted-up position, it means the handle **21** is fully lifted up, but the support **22** is just at a location between the work position and the transition position, which means the motherboard is not fully disconnected from the socket.
[0033] Then users need to pull the handle **21** horizontally, to move the support **22** towards the transition position (from FIG. **7** to FIG. **8**), and when the first pin **211** is at the right end of the second segment **111***b*, the support **22** arrives at the transition position, at this moment, the motherboard is fully disconnected from the socket, and the first pin **211** arrives at the bottom end of the third segment **111***c*.
[0034] Then users need to pull up the handle **21** and the support **22** together, to rotate the support **22** from the transition position to the open position, and during the rotation, the first pin **211** slides out of the top end of the third segment **111***c* (from FIG. **8** to FIG. **9**).
[0035] On the contrary, when the motherboard needs to be plugged back into the socket, users press down the handle **21** and the support **22** together, to rotate the support **22** from the open position to

the transition position, the first pin **211** will enter the top end of the third segment **111***c*, and then slide down to the bottom end of the third segment **111**.

[0036] Then users need to pull the handle **21** horizontally, to move the support **22** towards the work position, until the first pin **211** arrives at the left end of the second segment **111***b*. But at this moment the motherboard is not fully plugged into the socket.

[0037] Then users need to rotate down the handle **21** from the handle-lifted-up position to the handle-pressed-down position (from FIG. **7** to FIG. **6**), because the first pin **211** is limited in the first segment **111***a* and the arc groove **221** simultaneously, the first pin **211** pushes against the right-side of the first segment **111***a* shown in FIG. **6** and FIG. **7** during the rotation of the handle **21**, so that the support **22** is moved towards the work position. When the handle **21** arrives at the handle-pressed-down position, the support **22** arrives at the work position, at this moment, the motherboard is fully plugged into the socket.

[0038] In some embodiments, to avoid blocking the first pin **211**, there is a chamfer **111***d* between the first segment **111***a* and the second segment **111***b*, the chamfer **111***d* is used to prevent the first pin **211** from being stuck when entering the first segment **111***a* from the second segment **111***b*.

[0039] In some embodiments, the first pin **211** is a cylindrical pin, the width of the first segment **111***a*, the width of the second segment **111***b*, and the diameter of the first pin **211** are equal. The width of the third segment **111***c* is larger than the diameter of the first pin **211**, so that it is convenient for first pin **211** to enter or leave third segment **111***c*.

[0040] In some embodiments, as shown in FIG. **4** and FIG. **5**, the support **22** has a positioning hole **22***a*, the handle **21** has a latch **21***a* and a spring (not shown in figures). When the handle **21** arrives at the handle-lifted-up position, the latch **21***a* is aligned with the positioning hole **22***a*, the spring pushes the latch **21***a* into the positioning hole **22***a*, to position the handle **21** at the handle-lifted-up position, so that users can better push or pull the support **22** by the handle **21**. When users need to press down the handle **21** to the handle-pressed-down position, users need to pull out the latch **21***a* from the positioning hole **22***a* by hand, to unlock the handle **21** from the support **22**, then the handle **21** can be pressed down to the handle-pressed-down position.

[0041] As shown in FIG. **10** to FIG. **12**, in some embodiments, each second sheet **12** of the chassis **10** has a second sliding groove **121**, the support **22** has a second pin **222**, the second pin **222** is slidably inserted in the second sliding groove **121**. The second sliding groove **121** includes a straight-line segment **1211** and a positioning segment **1212**. The straight-line segment **1211** extends along the direction X, the positioning segment **1212** extends along the direction Y and is communicated to an end of the straight-line segment **1211**. When the support **22** moves between the work position and the transition position, the second pin **222** slides in the straight-line segment **1211**, so the second sheet **12** of the chassis **10** can support the support **22** by supporting the second pin **222**, which means the support **22** is not only supported by the bended sheet **11***a* mentioned-above, but also supported by the second sheet **12**. When the support **22** arrives at the transition position, the second pin **222** arrives at an top end of the positioning segment **1212**, and when the support **22** rotates from the transition position to the open position, due to the gravity and inertia of the support **22**, the second pin **222** will naturally fall into the positioning segment **1212**, and the positioning segment **1212** is used for positioning the second pin **222**, to stop the second pin **222** from moving along the direction X, so that the support **22** can rotate around the second pin **222** stably.

[0042] Furthermore, when the support **22** rotates from the open position back to the transition position, the second pin **222** needs to be moved from the positioning segment **1212** back to the straight-line segment **1211**, so as shown in FIG. **11**, the support **22** has a lever shaft **223**, the second sheet **12** of the chassis **10** has a supporting surface **12***a*, the supporting surface **12***a* is horizontal. When users press down the handle **21** to rotate the support **22** from the open position back to the transition position, the lever shaft **223** will push against the supporting surface **12***a* at a moment, then the support **22** can rotates around the lever shaft **223**, so the second pin **222** can be tilted up into the straight-line segment **1211** by the lever principle, then the second pin **222** can slide in the straight-line segment **1211** along the direction X, which allows the support **22** to move from the transition position back to the work position.

[0043] Furthermore, the second sliding groove **121** further includes an assembling segment **1213**, the assembling segment **1213** is communicated to the straight-line segment **1211** and extends through the top of the second sheet **12**. When assembling the second pin **222** into the straight-line segment **1211**, the second pin **222** passes through the assembling segment **1213** to the straight-line segment **1211**.

[0044] As shown in FIG. **13** to FIG. **15**, in some embodiments, the chassis **10** has a third sheet **13**, the third sheet **13** has a third sliding groove **131**, the third sliding groove **131** extends obliquely. The self-locking component **30** includes a spring sheet **31**, a link rod component **32**, and a third pin **33**, the spring sheet **31** is located on the third sheet **13** of the chassis **10**, the link rod component **32** connects the support **22** and the third pin **33**, the third pin **33** is slidably inserted in the third sliding groove **131**, the support **22** can slide the third pin **33** in the third sliding groove **131** by the link rod component **32**. The spring sheet **31** has a bevel **311** and a limiting hole **312**. When the support **22** rotates from the transition position to the open position, the third pin **33** will push against the bevel **311**, to deform the spring sheet **31** away from the third sheet **13**. When the third pin **33** arrives at the limiting hole **312**, the spring sheet **31** resets naturally, and the third pin **33** inserts into the limiting hole **312** simultaneously, then the third pin **33** is positioned in the limiting hole **312**, so the support **22** is positioned at the open position. When users need to rotate the support **22** from the open position back to the transition position, users need to pull the spring sheet **31** by hand, to deform the spring sheet **31** away from the third sheet **13**, until the third pin **33** is pulled out of the limiting hole **312**, then the support **22** can be rotated from the open position back to the transition position.

[0045] Furthermore, the link rod component **32** includes a first link rod **321**, a second link rod **322**, and a third link rod **323**. An end of the first link rod **321** connects an end of the second link rod **322**, another end of the first link rod **321** connects the support **22**, another end of the second link rod **322** connects the third sheet **13**, an end of the third link rod **323** connects the second link rod **322** at non-ends, another end of the third link rod **323** connects the third pin **33**. During the movement or the rotation of the support **22**, the end of the first link rod **321** connected to the support **22** must move and rotate with the support **22**, and the second link rod **322** must be rotated by the first link rod **321**, and the third link rod **323** must be moved and rotated by the second link rod **322**, and finally the third pin **33** must be moved in the third sliding groove **131**.

[0046] Furthermore, the spring sheet **31** includes a plane section **31***a* and a bump section **31***b*, the limiting hole **312** is defined on the plane section **31***a*, the plane section **31***a* is installed on the third sheet **13** by screws, the bump section **31***b* is arched relative to the plane section **31***a*, the bevel **311** is defined on the bump section **31***b*. When the support **22** rotates from the transition position to the open position, the third pin **33** pushes against the bevel **311** to deform the plane section **31***a* away from the third sheet **13**, the plane section **31***a* resets until the third pin **33** goes into the limiting hole **312**.

[0047] Furthermore, the plane section **31***a* has a notch **31***c*, the third sheet **13** of the chassis **10** has a guiding nail **132**, the guiding nail **132** includes a cylinder **1321** and a nail head **1322**, the cylinder **1321** extends through the notch **31***c* and is used to guide the deformation of the plane section **31***a*, the nail head **1322** is used to stop the deformation of the plane section **31***a* along the cylinder **1321**, to avoid the plane section **31***a* from breaking due to excessive deformation.

[0048] In some embodiments, the first sheet **11**, the second sheet **12**, the third sheet **13**, and the chassis **10** are formed by an integrated molding process, and thus, the first sliding groove **111**, the second sliding groove **121**, the supporting surface **12***a*, and the third sliding groove **131** are all defined on the chassis **10**.

[0049] In summary, the process of using cabinet **100** is: define an working state of cabinet **100** in which the support **22** is at the work position and the handle **21** is at the handle-pressed-down position, and the motherboard on the support **22** is plugged into the socket of the server **200** in the upper-layer space; when users need to maintain electronic devices in the lower-layer space, users hold the handle **21** and rotate the handle **21** from the handle-pressed-down position to the handle-lifted-up position, during the rotation of the handle **21**, the first pin **211** limited in the first segment **111***a* pushes against the first sheet **11** of the chassis **10**, so the support **22** is moved to the transition position, and when the handle **21** arrives at the handle-lifted-up position, the latch **21***a* inserts into the positioning hole **22***a*, to position the handle **21** on the handle-lifted-up position; after the handle **21** is positioned on the handle-lifted-up position, users need to pull the handle **21** horizontally along the direction X, to move the support **22** to the transition position; after the support **22** arrives at the transition position, users need to lift up the handle **21** to rotate the support **22** to the open position, at the moment the rotation begins, the second pin **222** moves into the positioning segment **1212** from the straight-line segment **1211**, and the first pin **211** slides out of the third segment **111***c*, until the support **22** arrives at the open position, the third pin **33** is positioned in the limiting hole **312** simultaneously, at this moment, the cabinet **100** is in an open state, so that users can maintain electronic devices in the lower-layer space easily.

[0050] When users need to put the cabinet **100** from the open state to the working state, users need to pull the plane section **31***a* of the spring sheet **31** by hand, to get the third pin **33** out of the limiting hole **312**, then user can press down the handle **21** to rotate the support **22** back to the transition position; during the rotation of the support **22** back to the transition position, the lever shaft **223** pushes against the supporting surface **12***a*, letting the support **22** rotates around the lever shaft **223**, so the second pin **222** can be lifted out of the positioning segment **1212** and goes into the straight-line segment **1211** by level principle; after the support **22** arrives at the transition position, users pushes the handle **21** horizontally along the direction X, to move the support **22** to the work position; when the first pin **211** arrives in the first segment **111***a*, users need to pull the latch **21***a* out of the positioning hole **22***a* by hand, to unlock the handle **21** relative to the support **22**, then press down the handle **21**, to rotate the handle **21** to the handle-pressed-down position; during the rotation of the handle **21** to the handle-pressed-down position, the first pin **211** limited in the first segment **111***a* pushes against the first sheet **11** of the chassis **10**, so that the support **22** is moved to the work position; when the handle **21** arrives at the handle-pressed-down position, the support **22** arrives at the work position, and the motherboard on the support **22** is plugged back into the socket of the server **200**.

[0051] The embodiments shown and described above are only examples. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, including in matters of shape, size, and arrangement of the parts within the principles of the present disclosure, up to and including the full extent established by the broad general meaning of the terms used in the claims.

Claims

1. A cabinet comprising: a chassis; a bracket comprising a support and a handle, wherein the support is movable between a work position and a transition position relative to the chassis, and the support is further rotatable between the transition position and an open position relative to the chassis, the handle is rotatable between a handle-lifted-up position and a handle-pressed-down position relative to the support; and a self-locking component connected to the support and the chassis, wherein when the handle rotates from the handle-pressed-down position to the handle-lifted-up position, the support moves from the work position to the transition position, when the support rotates from the transition position to the open position, the support extends out of the chassis, the self-locking component locks the support at the open position.

2. The cabinet of claim 1, wherein the chassis defines a first sliding groove and a second sliding groove, the support defines an arc groove, the handle comprises a first pin slidable in the arc groove, and the first pin is slidable into or out of the first sliding groove, the support comprises a second pin slidable in the second sliding groove, when the handle rotates from the handle-pressed-down position to the handle-lifted-up position, the first pin slides in the arc groove and slides in the first sliding groove simultaneously, the first pin pushes against the chassis to move the support from the work position to the transition position, and the second pin moves in the second sliding groove with the support; when the support rotates from the transition position to the open position, the support rotates around the second pin, and the first pin slides out of the first sliding groove; when the support rotates from the open position to the transition position, the support rotates around the second pin, the first pin slides into the first sliding groove; when the handle rotates from the handle-lifted-up position to the handle-pressed-down position, the first pin slides in the arc groove and slides in the first sliding groove simultaneously, the first pin pushes against the chassis to move the support from transition position to the work position, and the second pin moves in the second sliding groove with the support.
3. The cabinet of claim 2, wherein the second sliding groove comprises a straight-line segment and positioning segment, the straight-line segment extends along a direction of a movement of the support between the work position and the transition position, the positioning segment communicates to an end of the straight-line segment, when the support is at the transition position, the positioning segment is beneath the second pin, when the support rotates from the transition position to the open position, the second pin drops into the positioning segment to be positioned in the positioning segment.
4. The cabinet of claim 3, wherein the support further comprises a lever shaft, the chassis comprises a supporting surface, the supporting surface is parallel to the direction of the movement of the support between the work position and the transition position, when the support rotates from the open position to the transition position, the lever shaft pushes against the supporting surface to lever up the second pin, so that the second pin slides out of the positioning segment and slides into the straight-line segment.
5. The cabinet of claim 2, wherein the first sliding groove comprises a first segment, a second segment, and a third segment connected in sequence, the second segment extends in a direction of the movement of the support between the work position and the transition position, the first segment and the third segment communicate to two opposite sides of the second segment respectively, the third segment extends through a top side of the chassis so that the first pin can slide into or out of the first sliding groove, when the handle rotates between the handle-lifted-up position and the handle-pressed-down position, the first pin slides in the first segment and pushes against the chassis to move the support between the work position and the transition position; when the handle is at the handle-lifted-up position, the first pin is at the end of the second segment communicating to the first segment.
6. The cabinet of claim 5, wherein the support defines a positioning hole, the handle comprises a latch, when the handle is at the handle-lifted-up position, the latch is inserted into the positioning hole thereby positioning the handle at the handle-lifted-up position.
7. The cabinet of claim 1, wherein the self-locking component comprises a spring sheet, a link rod component, and a third pin, the spring sheet is located on the chassis, the link rod component connects the support and the third pin, the chassis defines a third sliding groove, the third pin is slidably inserted into the third sliding groove, the spring sheet defines a bevel and a limiting hole; the support moves the third pin by the link rod component, when the support moves to the handle-lifted-up position from the handle-pressed-down position, the third pin moves to the limiting hole and pushes the bevel to deform the spring sheet, when the third pin is aligned with the limiting hole, the spring sheet resets to insert the third pin into the limiting hole, so that the third pin is positioned in the limiting hole and the support is positioned at the handle-lifted-up position.

8. The cabinet of claim 7, wherein the link rod component comprises a first link rod, a second link rod, and a third link rod, the first link rod connects the second link rod and the support, the second link rod connects the first link rod and the chassis, the third link rod connects the second link rod and the third pin.

9. The cabinet of claim 7, wherein the spring sheet comprises a plane section and a bump section, the plane section is installed on the chassis, the limiting hole is located on the plane section, the bump section is arched relative to the plane section, the bevel is located on the bump section; when the third pin moves to the limiting hole, the third pin pushes the bevel to deform the plane section, until the third pin is aligned with the limiting hole, the plane section resets to insert the third pin into the limiting hole.

10. The cabinet of claim 9, wherein the plane section defines a notch, the chassis comprises a guiding nail, the guiding nail comprises a cylinder and a nail head, the cylinder extends through the notch to guide a direction of the deformation of the plane section, the nail head is configured for stopping the deformation of the plane section.

11. A server comprising: a plurality of electronic devices; and a cabinet comprising: a chassis configured for holding a part of the plurality of electronic devices; a bracket comprising a support and a handle, wherein the support is configured for holding another part of the plurality of electronic devices, the support is movable between a work position and a transition position relative to the chassis, and the support is further rotatable between the transition position and an open position relative to the chassis, the handle is rotatable between a handle-lifted-up position and a handle-pressed-down position relative to the support; and a self-locking component connected to the support and the chassis, wherein when the handle rotates from the handle-pressed-down position to the handle-lifted-up position, the support moves from the work position to the transition position, when the support rotates from the transition position to the open position, the support extends out of the chassis, the self-locking component locks the support at the open position.

12. The server of claim 11, wherein the chassis defines a first sliding groove and a second sliding groove, the support defines an arc groove, the handle comprises a first pin slidable in the arc groove, and the first pin is slidable into or out of the first sliding groove, the support comprises a second pin slidable in the second sliding groove, when the handle rotates from the handle-pressed-down position to the handle-lifted-up position, the first pin slides in the arc groove and slides in the first sliding groove simultaneously, the first pin pushes against the chassis to move the support from the work position to the transition position, and the second pin moves in the second sliding groove with the support; when the support rotates from the transition position to the open position, the support rotates around the second pin, and the first pin slides out of the first sliding groove; when the support rotates from the open position to the transition position, the support rotates around the second pin, the first pin slides into the first sliding groove; when the handle rotates from the handle-lifted-up position to the handle-pressed-down position, the first pin slides in the arc groove and slides in the first sliding groove simultaneously, the first pin pushes against the chassis to move the support from transition position to the work position, and the second pin moves in the second sliding groove with the support.

13. The server of claim 12, wherein the second sliding groove comprises a straight-line segment and positioning segment, the straight-line segment extends along a direction of a movement of the support between the work position and the transition position, the positioning segment communicates to an end of the straight-line segment, when the support is at the transition position, the positioning segment is beneath the second pin, when the support rotates from the transition position to the open position, the second pin drops into the positioning segment to be positioned in the positioning segment.

14. The server of claim 13, wherein the support further comprises a lever shaft, the chassis comprises a supporting surface, the supporting surface is parallel to the direction of the movement of the support between the work position and the transition position, when the support rotates from the open position to the transition position, the lever shaft pushes against the supporting surface to lever up the second pin, so that the second pin slides out of the positioning segment and slides into the straight-line segment.

15. The server of claim 12, wherein the first sliding groove comprises a first segment, a second segment, and a third segment connected in sequence, the second segment extends in a direction of the movement of the support between the work position and the transition position, the first segment and the third segment communicate to two opposite sides of the second segment respectively, the third segment extends through a top side of the chassis so that the first pin can slide into or out of the first sliding groove, when the handle rotates between the handle-lifted-up position and the handle-pressed-down position, the first pin slides in the first segment and pushes against the chassis to move the support between the work position and the transition position; when the handle is at the handle-lifted-up position, the first pin is at the end of the second segment communicating to the first segment.

16. The server of claim 11, wherein the self-locking component comprises a spring sheet, a link rod component, and a third pin, the spring sheet is located on the chassis, the link rod component connects the support and the third pin, the chassis defines a third sliding groove, the third pin is slidably inserted into the third sliding groove, the spring sheet defines a bevel and a limiting hole; the support moves the third pin by the link rod component, when the support moves to the handle-lifted-up position from the handle-pressed-down position, the third pin moves to the limiting hole and pushes the bevel to deform the spring sheet, when the third pin is aligned with the limiting hole, the spring sheet resets to insert the third pin into the limiting hole, so that the third pin is positioned in the limiting hole and the support is positioned at the handle-lifted-up position.

17. The server of claim 16, wherein the link rod component comprises a first link rod, a second link rod, and a third link rod, the first link rod connects the second link rod and the support, the second link rod connects the first link rod and the chassis, the third link rod connects the second link rod and the third pin.

18. The server of claim 16, wherein the spring sheet comprises a plane section and a bump section, the plane section is installed on the chassis, the limiting hole is located on the plane section, the bump section is arched relative to the plane section, the bevel is located on the bump section; when the third pin moves to the limiting hole, the third pin pushes the bevel to deform the plane section, until the third pin is aligned with the limiting hole, the plane section resets to insert the third pin into the limiting hole.

19. The server of claim 18, wherein the plane section defines a notch, the chassis comprises a guiding nail, the guiding nail comprises a cylinder and a nail head, the cylinder extends through the notch to guide a direction of the deformation of the plane section, the nail head is configured for stopping the deformation of the plane section.

20. A cabinet comprising: a chassis defining a lower-layer space and an upper-layer space, the chassis configured for holding electronic devices in the lower-layer space; a bracket comprising a support and a handle, wherein the support is configured for holding electronic devices in the upper-layer space, the support is movable between a work position and a transition position relative to the chassis, and the support is further rotatable between the transition position and an open position relative to the chassis, the handle is rotatable between a handle-lifted-up position and a handle-pressed-down position relative to the support; and a self-locking component connected to the support and the chassis, wherein when the handle rotates from the handle-pressed-down position to the handle-lifted-up position, the support moves from the work position to the transition position, when the support rotates from the transition position to the open position, the support extends out of the upper-layer space to expose the lower-layer space, the self-locking component locks the support at the open position.